

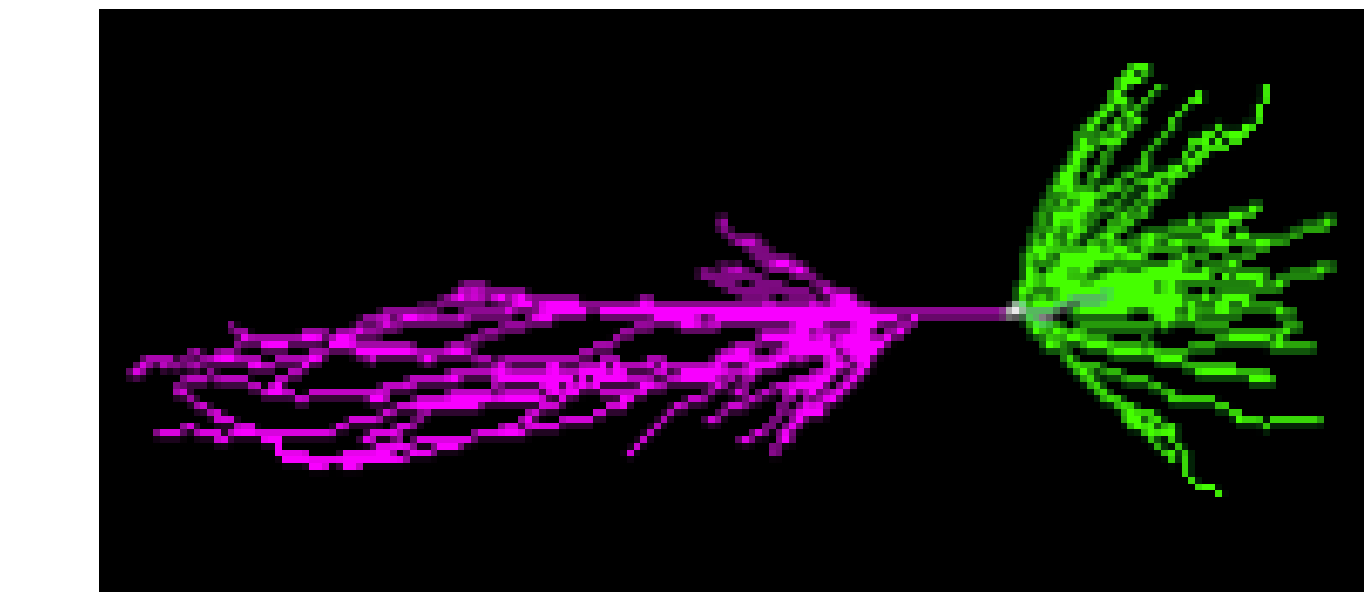
A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

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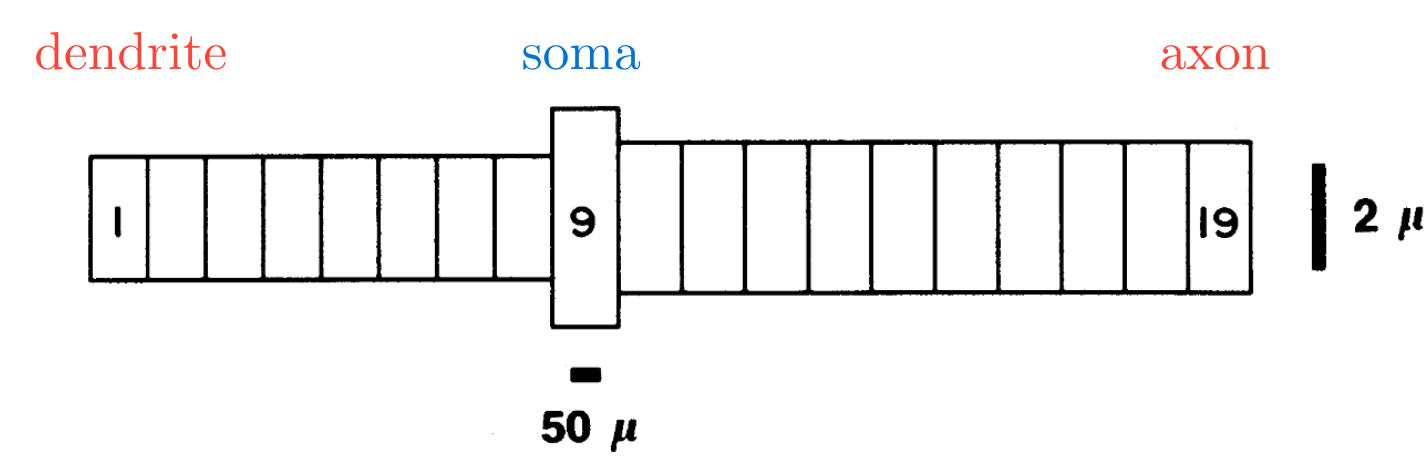
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Introduction

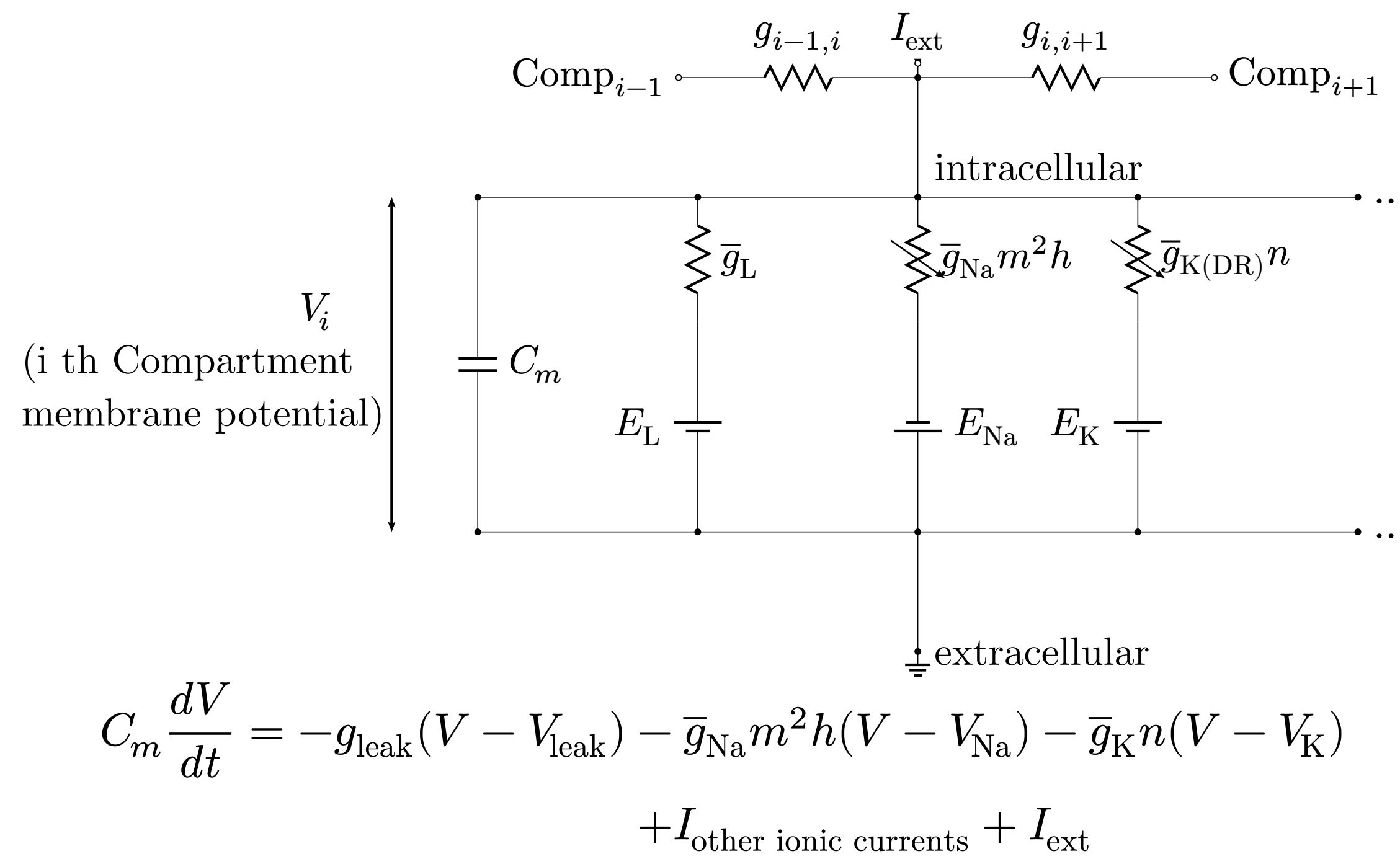


Guinea pig CA3 pyramidal neuron[1]



Modeled as

19 Compartments (comps) [2]



The conductances depend on **gate variables**, which follow ODEs with rate functions α, β :

$$I_{Na} = \bar{g}_{Na} m^2 h (V - E_{Na})$$

$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$

$$\alpha_m(V) = 0.32(13.1 - u)/(\exp((13.1 - u)/4) - 1)$$

$$\beta_m(V) = 0.28(u - 40.1)/(\exp((u - 40.1)/5) - 1)$$

→ **11 states (10 gates and V) per comp**

→ **209 states** for 19 comps;

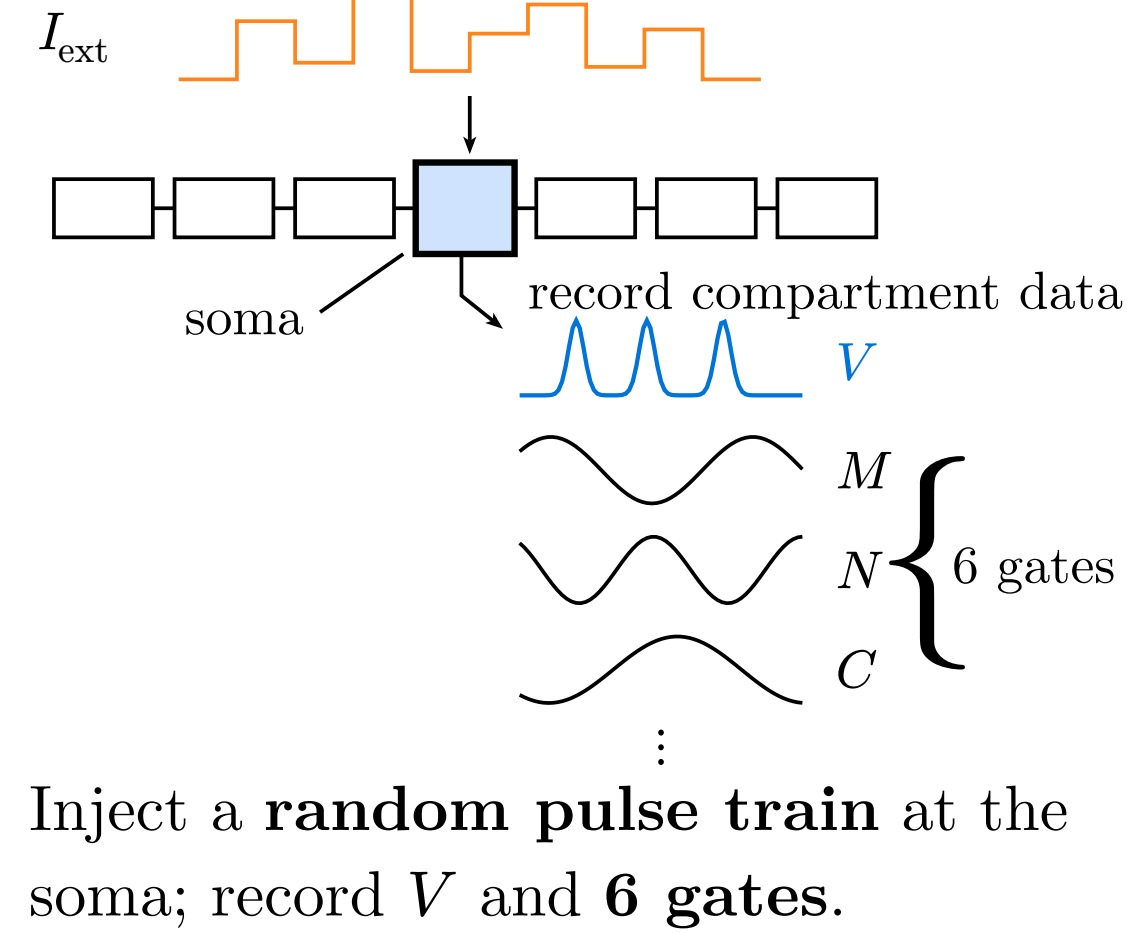
In large scale network simulations, the number of gate variables becomes a **memory bottleneck**.

Purpose

Development of a multi-compartment neuron surrogate model capable of reproducing the membrane potential response with fewer gate variables.

Methods

① Sample training data



Inject a **random pulse train** at the soma; record V and **6 gates**.

② Compress the gates

